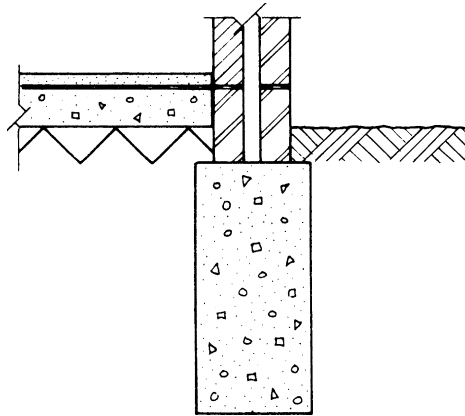


4 SUBSTRUCTURE



FOUNDATIONS – FUNCTION, MATERIALS AND SIZING

FOUNDATION BEDS

SHORT BORED PILE FOUNDATIONS

FOUNDATION TYPES AND SELECTION

PILED FOUNDATIONS

RETAINING WALLS

GABIONS AND MATTRESSES

BASEMENT CONSTRUCTION

WATERPROOFING BASEMENTS

EXCAVATIONS

CONCRETE PRODUCTION

COFFERDAMS

STEEL SHEET PILING

CAISSONS

UNDERPINNING

GROUND WATER CONTROL

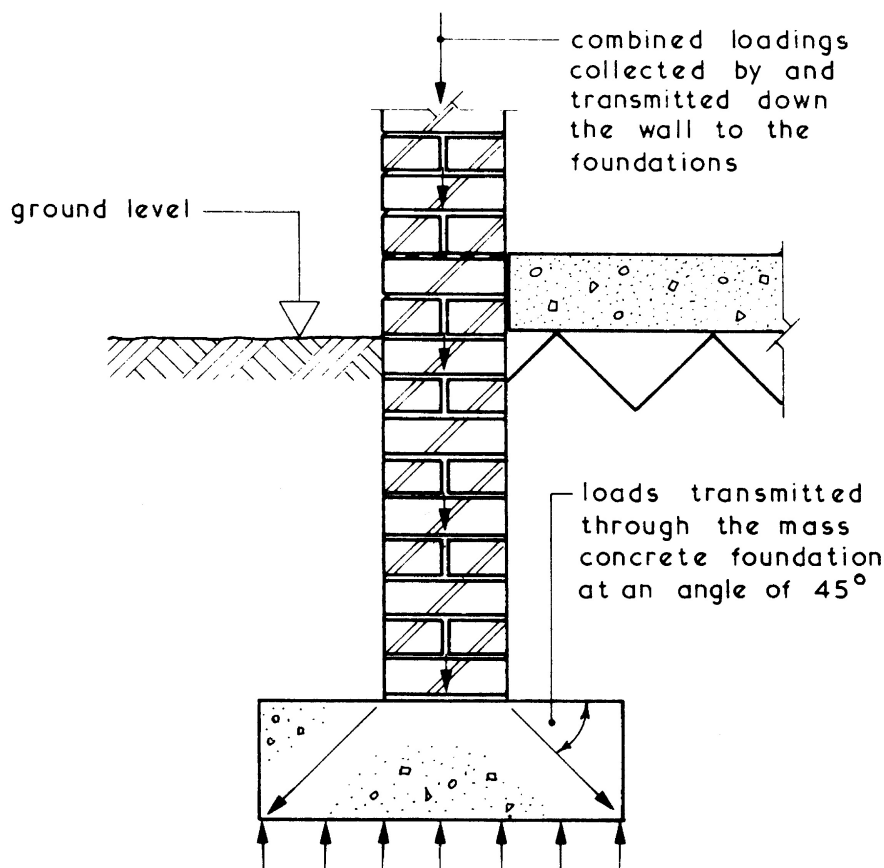
SOIL STABILISATION AND IMPROVEMENT

RECLAMATION OF WASTE LAND

TREATMENT OF CONTAMINATED SUBSOIL

Foundations ~ the function of any foundation is to safely sustain and transmit to the ground on which it rests the combined dead, imposed and wind loads in such a manner as not to cause any settlement or other movement which would impair the stability or cause damage to any part of the building.

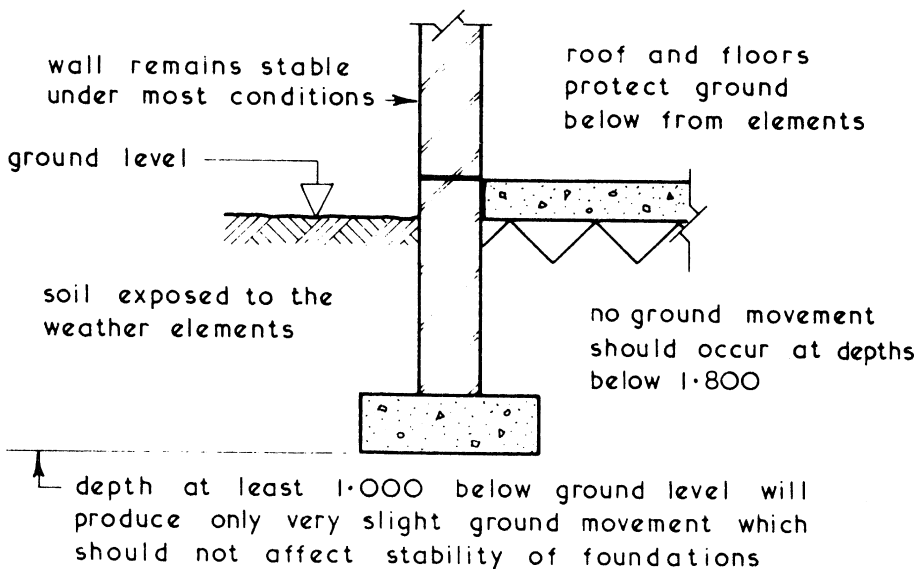
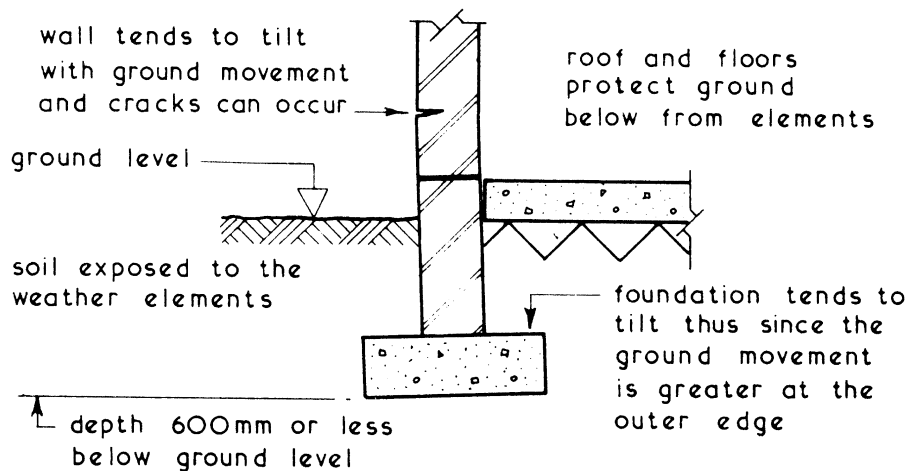
Example ~



Subsoil beneath foundation is compressed and reacts by exerting an upward pressure to resist foundation loading. If foundation load exceeds maximum passive pressure of ground (i.e. bearing capacity) a downward movement of the foundation could occur. Remedy is to increase plan size of foundation to reduce the load per unit area or alternatively reduce the loadings being carried by the foundations.

Subsoil Movements ~ these are due primarily to changes in volume when the subsoil becomes wet or dry and occurs near the upper surface of the soil. Compact granular soils such as gravel suffer very little movement whereas cohesive soils such as clay do suffer volume changes near the upper surface. Similar volume changes can occur due to water held in the subsoil freezing and expanding – this is called Frost Heave.

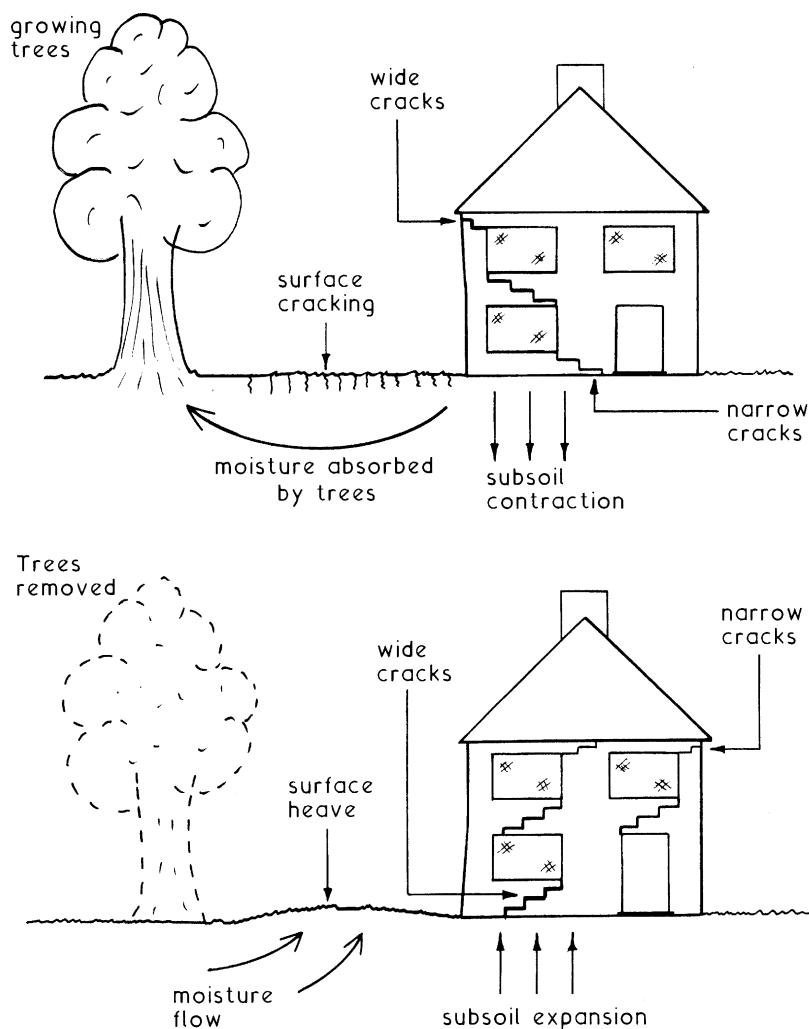
Typical Examples ~



Foundations – Subsoil Movements

Trees ~ damage to foundations. Substructural damage to buildings can occur with direct physical contact by tree roots. More common is the indirect effect of moisture shrinkage or heave, particularly apparent in clay subsoils.

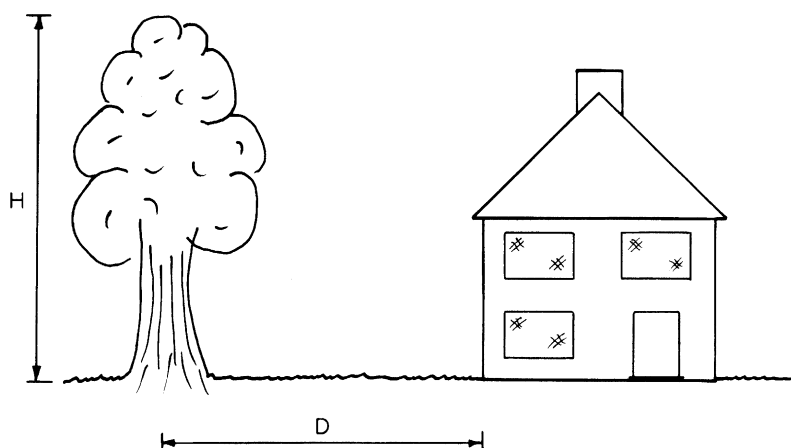
Shrinkage is most evident in long periods of dry weather, compounded by moisture abstraction from vegetation. Notably broad-leaved trees such as oak, elm and poplar in addition to the thirsty willow species. Heave is the opposite. It occurs during wet weather and is compounded by previous removal of moisture-dependent trees that would otherwise effect some drainage and balance to subsoil conditions.



Trees ~ effect on foundations. Trees up to 30 m distance may have an effect on foundations, therefore reference to local authority building control policy should be undertaken before specifying construction techniques.

Traditional strip foundations are practically unsuited, but at excavation depths up to 2.5 or 3.0m, deep strip or trench fill (preferably reinforced) may be appropriate. Short bored pile foundations are likely to be more economical and particularly suited to depths exceeding 3.0m.

For guidance only, the illustration and table provide an indication of foundation depths in shrinkable subsoils.



H = Mature height of tree
 D = Distance to centre of tree

D/H - Distance from tree/Height of tree

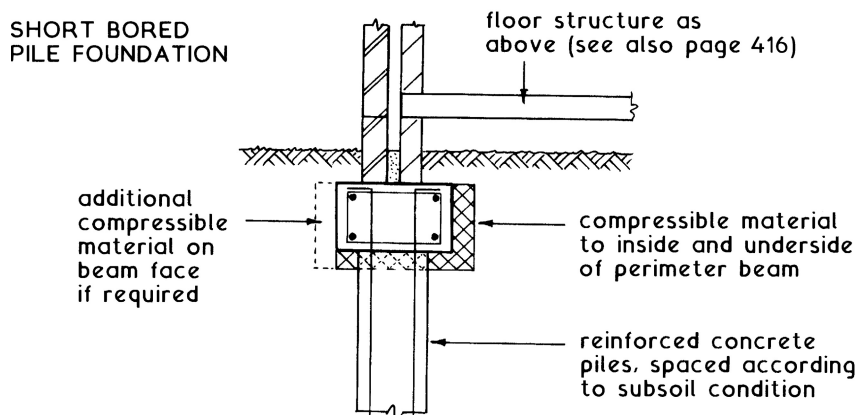
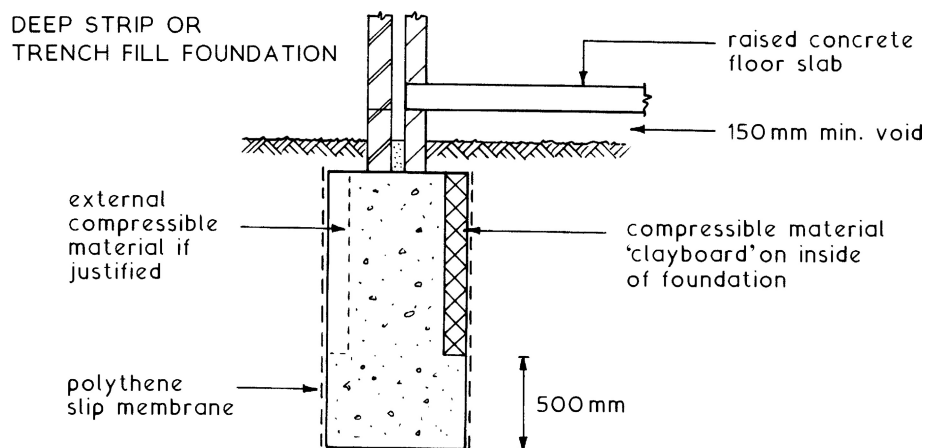
Tree species	0.10	0.25	0.33	0.50	0.66	0.75	1.00
Oak, elm, poplar and willow	3.00	2.80	2.60	2.30	2.10	1.90	1.50
All others	2.80	2.40	2.10	1.80	1.50	1.20	1.00

Minimum foundation depth (m)

Foundations – Subsoil Movements

Trees ~ preservation orders (see page 155) may be waived by the local planning authority. Permission for tree felling is by formal application and will be considered if the proposed development is in the economic and business interests of the community. However, tree removal is only likely to be acceptable if there is an agreement for replacement stock being provided elsewhere on the site.

In these circumstances there is potential for ground heave within the 'footprint' of felled trees. To resist this movement, foundations must incorporate an absorbing layer or compressible filler with ground floor suspended above the soil.



Cracking in Walls ~ cracks are caused by applied forces which exceed those that the building can withstand. Most cracking is superficial, occurring as materials dry out and subsequently shrink to reveal minor surface fractures of <2mm. These insignificant cracks can be made good with proprietary fillers.

Severe cracking in walls may result from foundation failure, due to inadequate design or physical damage. Further problems could include:

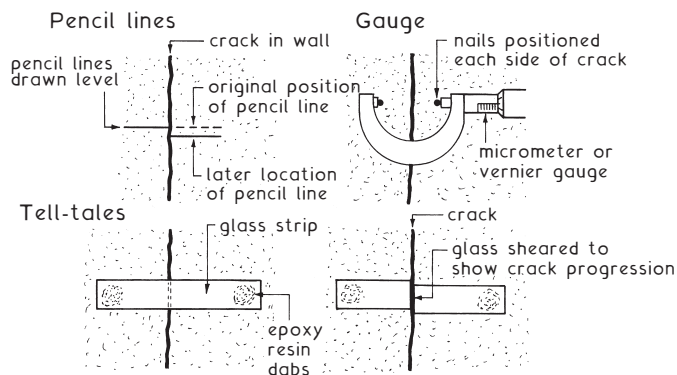
- * Structural instability
- * Air infiltration
- * Sound insulation reduction
- * Rain penetration
- * Heat loss
- * Visual depreciation

A survey should be undertaken to determine:

1. The cause of cracking, i.e.
 - * Loads applied externally (tree roots, subsoil movement).
 - * Climate/temperature changes (thermal movement).
 - * Moisture content change (faulty dpc, building leakage).
 - * Vibration (adjacent work, traffic).
 - * Changes in physical composition (salt or ice formation).
 - * Chemical change (corrosion, sulphate attack).
 - * Biological change (timber decay).
2. The effect on a building's performance (structural and environmental).
3. The nature of movement – completed, ongoing or intermittent (seasonal).

Observations over a period of several months, preferably over a full year, will determine whether the cracking is new or established and whether it is progressing.

Simple method for monitoring cracks –



Further reading – BRE Digest 251: Assessment of damage in low rise buildings.

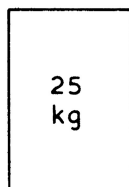
Foundation Materials ~ from page 248 one of the functions of a foundation can be seen to be the ability to spread its load evenly over the ground on which it rests. It must of course be constructed of a durable material of adequate strength. Experience has shown that the most suitable material is concrete.

Concrete is a mixture of cement + aggregates + water in controlled proportions.

CEMENT

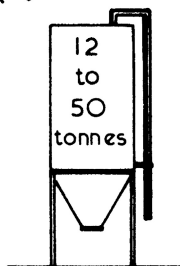
Manufactured from clay and chalk and is the matrix or binder of the concrete mix. Cement powder can be supplied in bags or bulk –

Bags ~



airtight sealed bags requiring a dry, damp-free store.

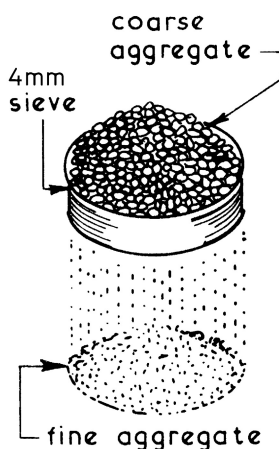
Bulk ~



delivered by tanker and pumped into storage silo.

AGGREGATES

Coarse aggregate is generally defined as a material which is retained on a 4mm sieve.



Fine aggregate is generally defined as a material which passes a 4mm sieve. Aggregates can be either natural rock which has disintegrated or crushed stone or gravel.

WATER

Must be of a quality fit for drinking.

MIXES

These are expressed as a ratio thus:

1:3:6/20mm

which means –
1 part cement.

3 parts of fine aggregate.

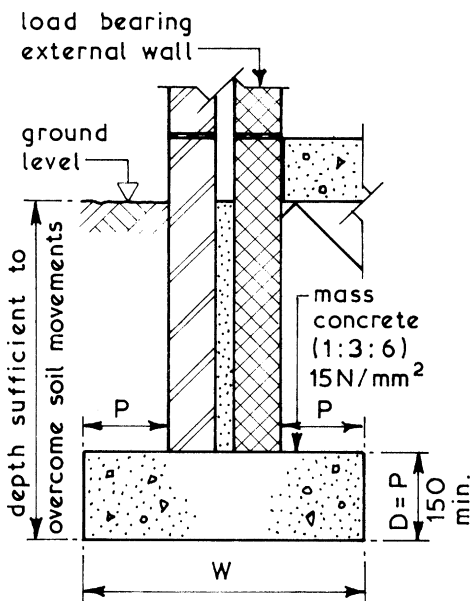
6 parts of coarse aggregate.

20mm – maximum size of coarse aggregate for the mix.

Water is added to start the chemical reaction and to give the mix workability ~ the amount used is called the

Water/Cement Ratio and is usually about 0.4 to 0.5.

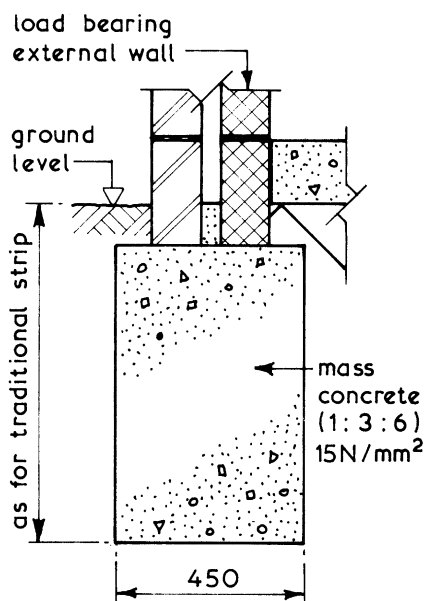
Too much water will produce a weak concrete of low strength whereas too little water will produce a concrete mix of low and inadequate workability.



$$W = \frac{\text{load per metre}}{\text{bearing capacity of soil or}}$$

W = not less than that given in Table on page 259

NB. In all cases W must give adequate working space which is usually 450 to 600 mm minimum depending on depth of excavation.

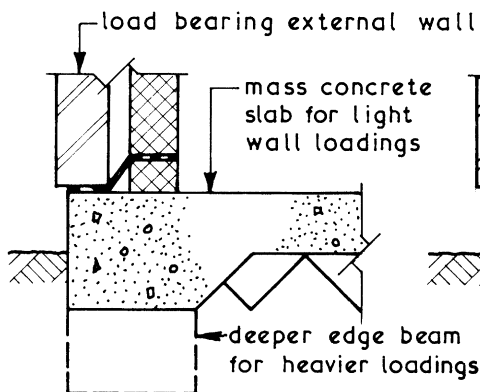


generally considered to be cheaper than traditional strip foundations since:

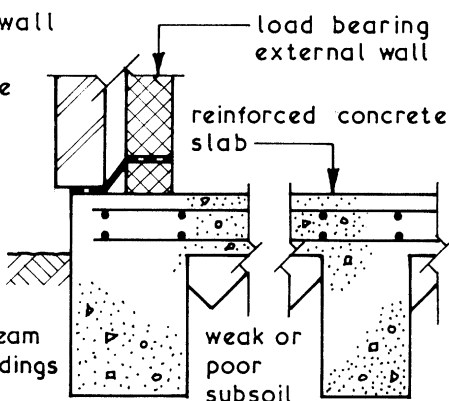
1. fewer man hours required.
2. requires less skilled trades.
3. uses ready-mix concrete, therefore less material is stored on site making it clearer and easier to manage.

TRADITIONAL STRIP

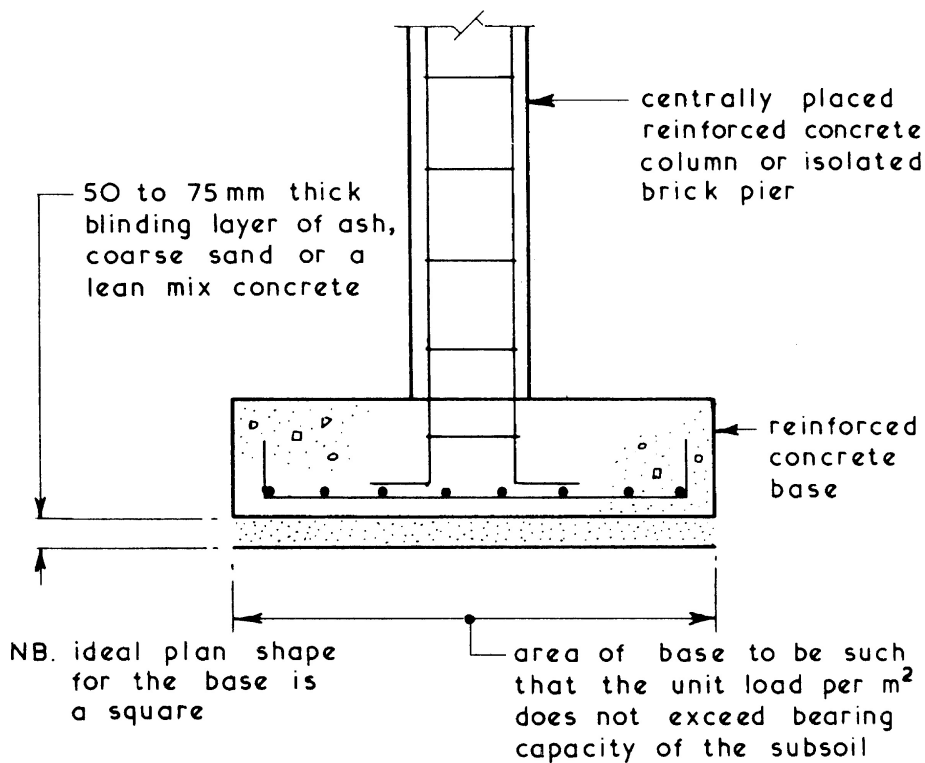
DEEP STRIP OR TRENCH FILL



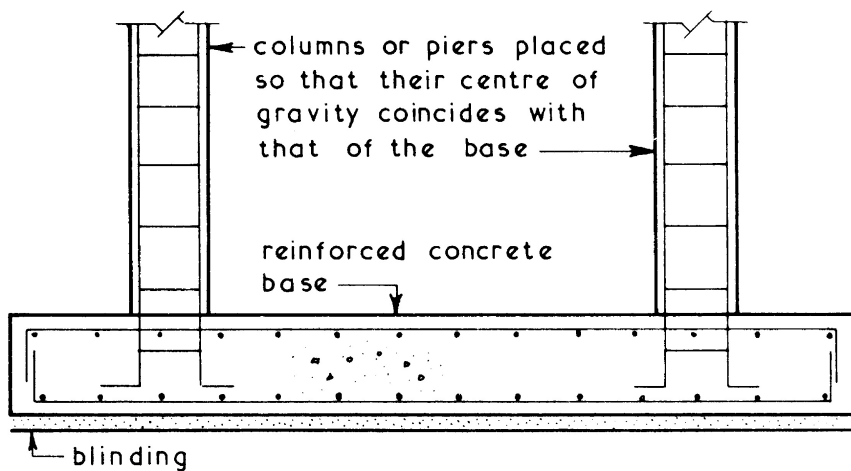
SOLID SLAB RAFT



BEAM AND SLAB RAFT



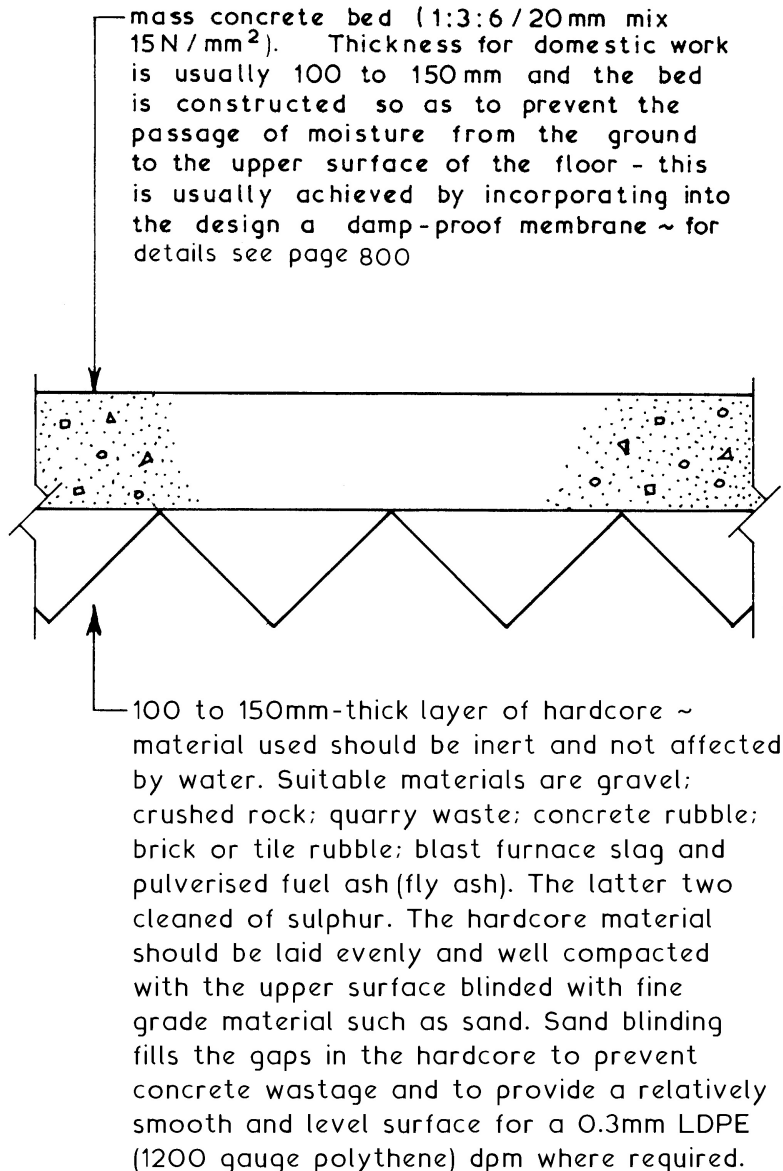
ISOLATED PAD



COMBINED PAD

Bed ~ a concrete slab resting on and supported by the subsoil, usually forming the ground floor surface. Beds (sometimes called over-site concrete) are usually cast on a layer of hardcore which is used to make up the reduced level excavation and thus raise the level of the concrete bed to a position above ground level.

Typical Example ~

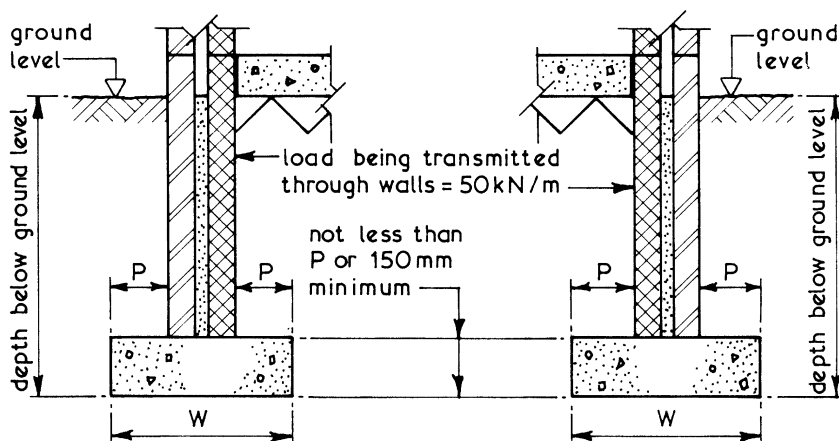


Basic Sizing ~ the size of a foundation is basically dependent on two factors:

1. Load being transmitted, max 70kN/m (dwellings up to three storeys).
2. Bearing capacity of subsoil under proposed foundation.

For guidance on bearing capacities for different types of subsoil see BS EN 1997-1: Geotechnical design. General rules and BS 8103-1: Structural design of low rise buildings. Also, directly from soil investigation results.

Typical Examples ~



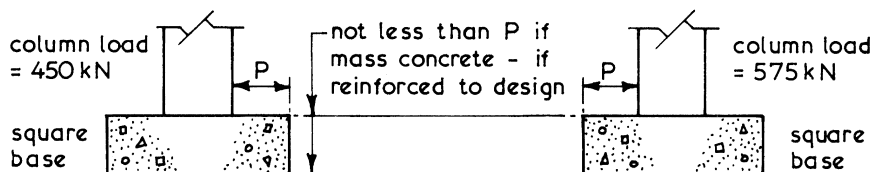
safe bearing capacity of compact gravel subsoil = 100 kN/m²

$$W = \frac{\text{load}}{\text{bearing capacity}} = \frac{50}{100} = 500 \text{ mm minimum}$$

safe bearing capacity of clay subsoil = 80 kN/m²

$$W = \frac{\text{load}}{\text{bearing capacity}} = \frac{50}{80} = 625 \text{ mm minimum}$$

The above widths may not provide adequate working space within the excavation and can be increased to give required space. Guidance on the minimum width for a limited range of applications can be taken from the table on the next page.



bearing capacity of subsoil 150 kN/m²

$$\text{area of base} = \frac{\text{load}}{\text{bc}} = \frac{450}{150}$$

$$= 3 \text{ m}^2 \therefore \text{side} = \sqrt{3} = 1.732 \text{ min.}$$

bearing capacity of subsoil 85 kN/m²

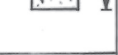
$$\text{area of base} = \frac{\text{load}}{\text{bc}} = \frac{575}{85}$$

$$= 6.765 \text{ m}^2 \therefore \text{side} = \sqrt{6.765} = 2.6 \text{ min.}$$

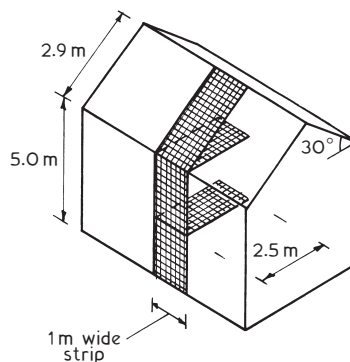
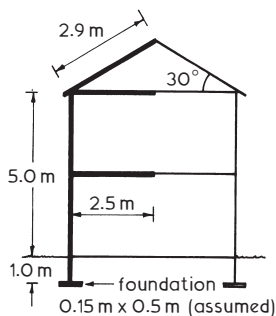
Ground type	Ground condition	Field test	Max. total load on load-bearing wall (kN/m)									
			20	30	40	50	60	70				
			Minimum width (mm)									
Rock	Not inferior to sandstone, limestone or firm chalk	Requires a mechanical device to excavate.	At least equal to the width of the wall									
Gravel	Medium density	Pick required to excavate.										
Sand	Compact	50 mm square peg hard to drive beyond 150 mm.	250	300	400	500	600	650				
Clay	Stiff	Requires pick or mechanical device to aid removal. Can be indented slightly with thumb.										
Sandy clay	Stiff		250	300	400	500	600	650				
Clay	Firm	Can be moulded under substantial pressure by fingers.										
Sandy clay	Firm		300	350	450	600	750	850				
Sand	Loose	Can be excavated by spade. 50 mm square peg easily driven.										
Silty sand	Loose											
Clayey sand	Loose		400	600	Conventional strip foundations unsuitable for a total load exceeding 30 kN/m							
Silt	Soft	Finger pushed in up to 10 mm. Easily moulded with fingers.										
Clay	Soft		450	650								
Sandy clay	Soft											
Silty clay	Soft											
Silt	Very soft	Finger easily pushed in up to 25 mm. Wet sample exudes between fingers when squeezed.	Conventional strip inappropriate.									
Clay	Very soft		Steel-reinforced wide strip, deep strip or piled foundation selected subject to specialist advice									
Sandy clay	Very soft											
Silty clay	Very soft											

Adapted from Table 10 in the Bldg. Regs., A.D: A – Structure.

Adapted from Table 10 in the Bldg. Regs., A.D: A – Structure.

	20	30	40	50	60	70 kN/m max. load
Rock	As described on the preceding page					
Gravel Sand (Medium density, compact)						
Clay Sandy clay (Stiff)						
Clay Sandy clay (Firm)						
Sand Silty sand Clayey sand (Loose)			Strip foundation requirements not within the criteria defined by the guidance in Approved Document A. Therefore, specific design will be required where the building Load is greater than 30 kN per metre run of wall. Note: Foundation dimensions in millimetres.			
Silt clay Sandy clay Silty clay (Soft)						
Silt clay Sandy clay Silty clay (Very soft)						

Typical Procedure (for guidance only) –



Dead load per m run (see pages 39 and 40)

Substructure brickwork, 1m × 1m × 476 kg/m ²	=	476 kg
..... cavity conc. (50 mm), 1m × 1m × 2300 kg/m ³	=	115 kg
Foundation concrete, 0.15 m × 1m × 0.5 m × 2300 kg/m ³	=	173 kg
Superstructure brickwork, 5 m × 1m × 221 kg/m ²	=	1105 kg
..... blockwork & ins., 5 m × 1m × 79 kg/m ²	=	395 kg
..... 2 coat plasterwork, 5 m × 1m × 22 kg/m ²	=	110 kg
Floor joists/boards/plstrbrd., 2.5 m × 1m × 42.75 kg/m ²	=	107 kg
Ceiling joists/plstrbrd/ins., 2.5 m × 1m × 19.87 kg/m ²	=	50 kg
Rafters, battens & felt, 2.9 m × 1m × 12.10 kg/m ²	=	35 kg
Single lap tiling, 2.9 m × 1m × 49 kg/m ²	=	142 kg
		2708 kg

Note: kg × 9.81 = Newtons

Therefore: 2708 kg × 9.81 = 26565 N or 26.56 kN

Imposed load per m run (see BS EN 1991-1-1: Densities, self-weight, imposed loads for buildings) –

Floor, 2.5 m × 1m × 1.5 kN/m² = 3.75 kN

Roof, 2.9 m × 1m × 1.5 kN/m² (snow) = 4.05 kN

7.80 kN

Note: For roof pitch >30°, snow load = 0.75 kN/m²

Dead + imposed load is 26.56 kN + 7.80 kN = 34.36 kN

Given that the subsoil has a safe bearing capacity of 75 kN/m²,

W = load ÷ bearing capacity = 34.36 ÷ 75 = 0.458 m or 458 mm

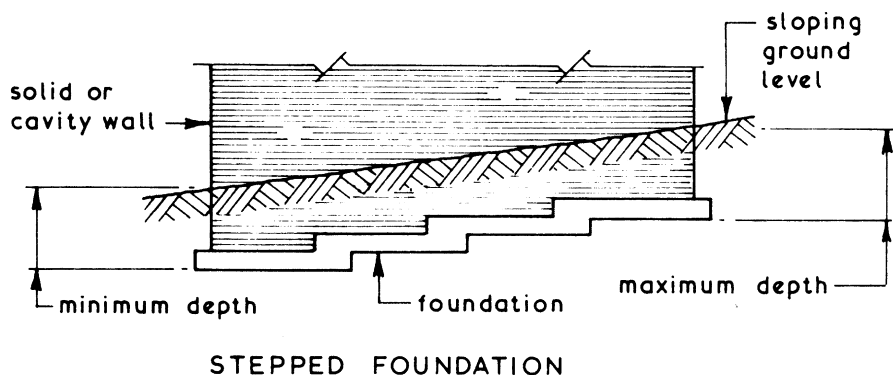
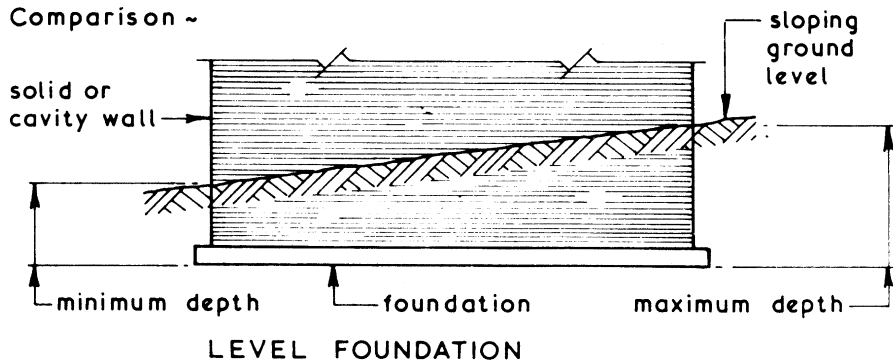
Therefore a foundation width of 500 mm is adequate.

Note: This example assumes the site is sheltered. If it is necessary to make allowance for wind loading, reference should be made to BS EN 1991-1-4: Wind actions.

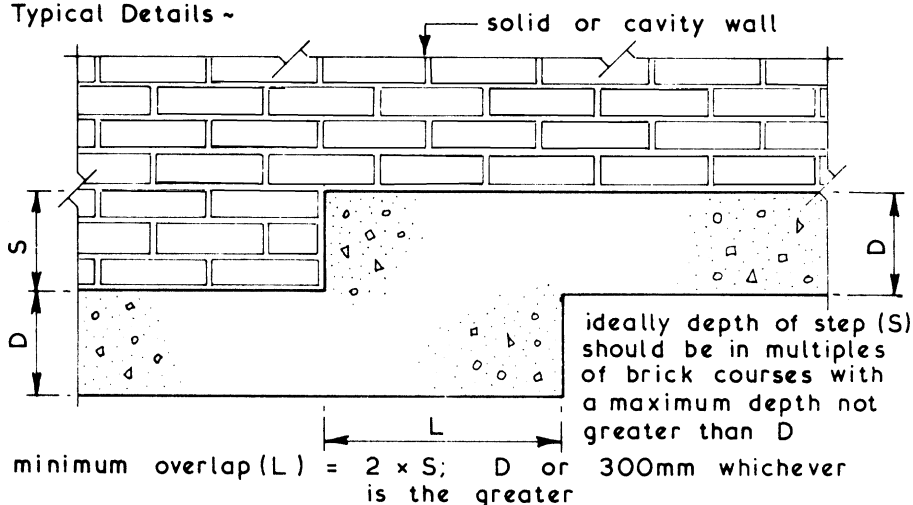
Stepped Foundations

Stepped Foundations ~ these are usually considered in the context of strip foundations and are used mainly on sloping sites to reduce the amount of excavation and materials required to produce an adequate foundation.

Comparison ~

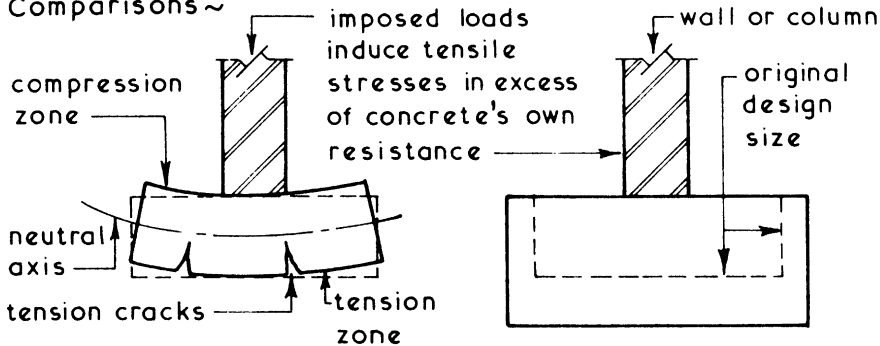


Typical Details ~



Concrete Foundations ~ concrete is a material which is strong in compression but weak in tension. If its tensile strength is exceeded cracks will occur, resulting in a weak and unsuitable foundation. One method of providing tensile resistance is to include in the concrete foundation bars of steel as a form of reinforcement to resist all the tensile forces induced into the foundation. Steel is a material which is readily available and has high tensile strength.

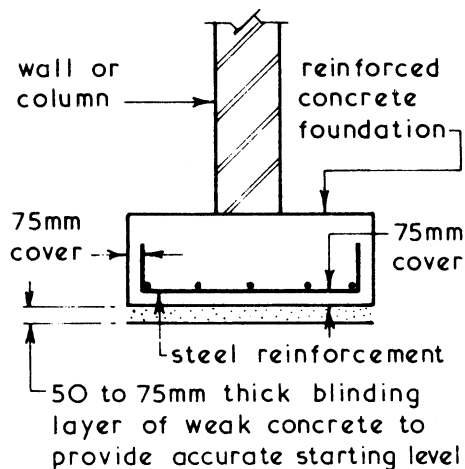
Comparisons ~



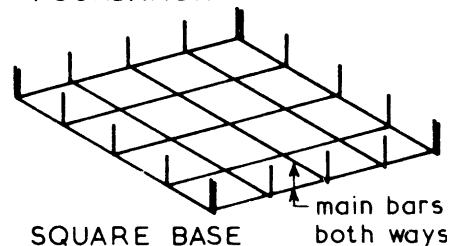
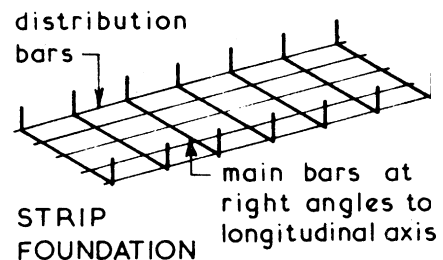
foundation tends to bend, the upper fibres being compressed and the lower fibres being stretched and put in tension - remedies: increase size of base or design as a reinforced concrete foundation

size of foundation increased to provide the resistance against the induced tensile stresses - generally not economic due to the extra excavation and materials required

Typical RC Foundation

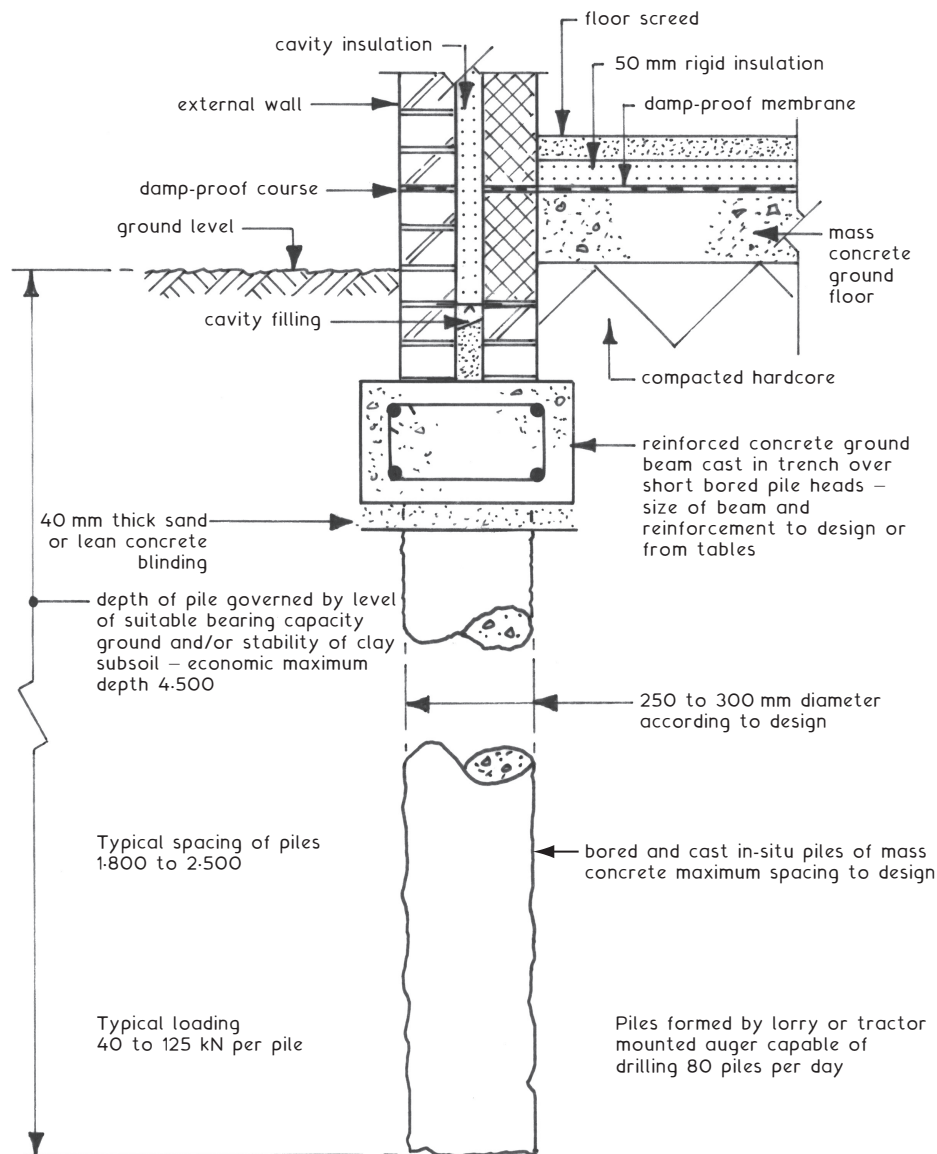


Reinforcement Patterns

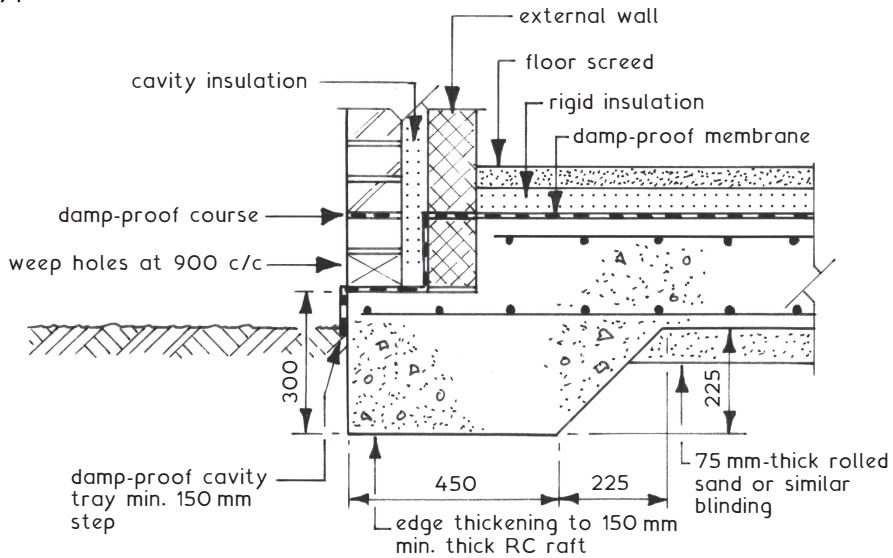


Short Bored Piles ~ these are a form of foundation which are suitable for domestic loadings and clay subsoils where ground movements can occur below the 1.000 depth associated with traditional strip and trench fill foundations. They can be used where trees are planted close to a new building since the trees may eventually cause damaging ground movements due to extracting water from the subsoil and root growth. Conversely where trees have been removed this may lead to ground swelling.

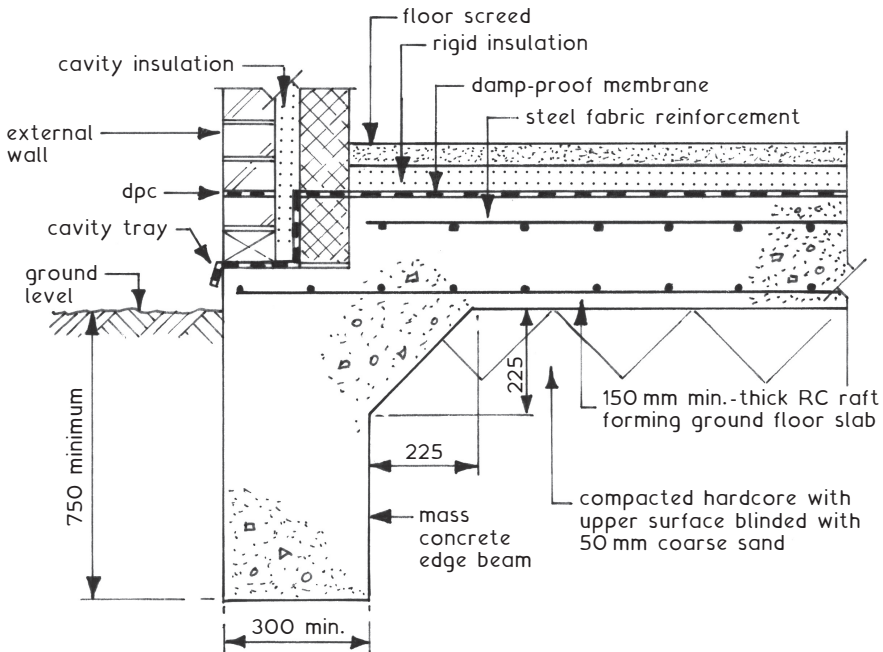
Typical Details ~



Typical Details ~



REINFORCED CONCRETE RAFT WITH EDGE THICKENING

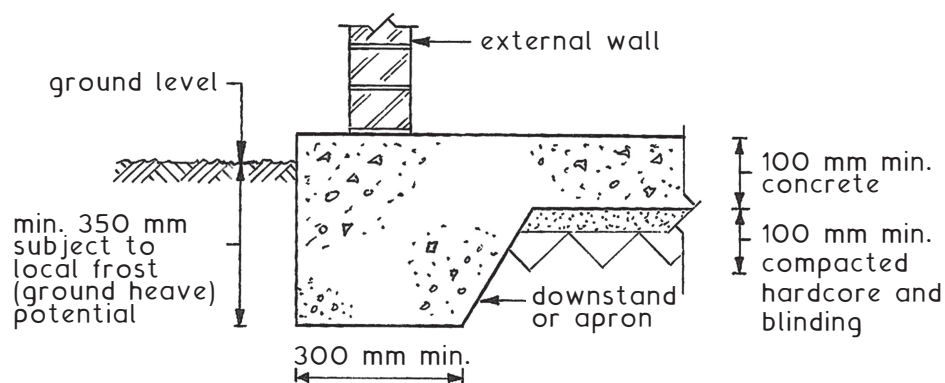


REINFORCED CONCRETE RAFT WITH EDGE BEAM

Domestic Garages and Other Non-Habitable Buildings ~ buildings with relatively light structural loading on their peripheral walls (typically single storey storage facilities), can have less stringent construction requirements than habitable accommodation. Building Regulations must still be adhered to, not least with construction of foundations that will adequately support all dead and imposed loads with regard to the properties and bearing capacity of the ground.

Integral and Attached Domestic Garages ~ to maintain continuity foundations should be the same as specified for the house, thereby reducing the possibility of differential movement. Also, if the garage is single storey, potential exists for constructing above without the need for new foundations.

Detached Garages ~ if subsoil quality is poor, an edge thickened steel mesh reinforced slab as shown on the previous page will be required. Otherwise, an unreinforced edge thickened concrete slab will be sufficient.



Garage Floor Slab ~ shrinkage control is required for concrete exceeding 16 m² in area. Steel mesh reinforcement is usual, but provision of a movement joint central to the slab is an alternative.

Foundation Design Principles ~ the main objectives of foundation design are to ensure that the structural loads are transmitted to the subsoil(s) safely, economically and without any unacceptable movement during the construction period and throughout the anticipated life of the building or structure.

Basic Design Procedure ~ this can be considered as a series of steps or stages:

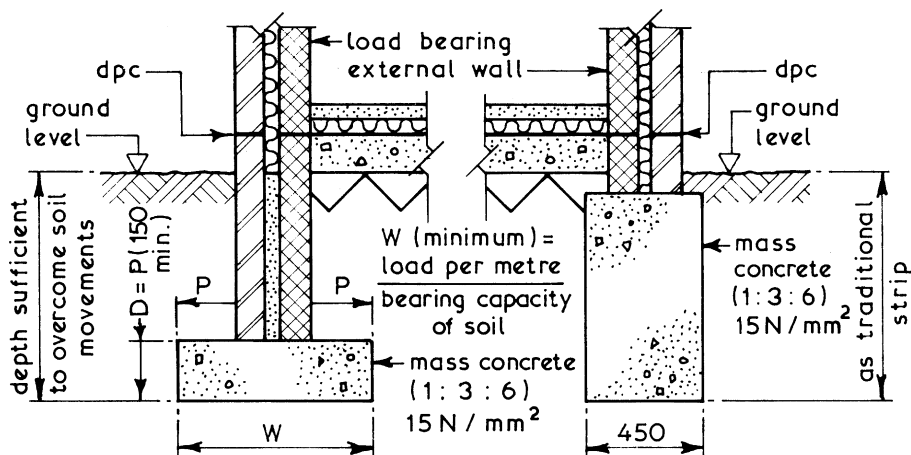
1. Assessment of site conditions in the context of the site and soil investigation report.
2. Calculation of anticipated structural loading(s).
3. Choosing the foundation type taking into consideration:
 - Soil conditions;
 - Type of structure;
 - Structural loading(s);
 - Economic factors;
 - Time factors relative to the proposed contract period;
 - Construction problems.
4. Sizing the chosen foundation in the context of loading(s), ground bearing capacity and any likely future movements of the building or structure.

Foundation Types ~ apart from simple domestic foundations most foundation types are constructed in reinforced concrete and may be considered as being shallow or deep. Most shallow types of foundation are constructed within 2·000 of the ground level but in some circumstances it may be necessary to take the whole or part of the foundations down to a depth of 2·000 to 5·000 as in the case of a deep basement where the structural elements of the basement are to carry the superstructure loads. Generally foundations which need to be taken below 5·000 deep are cheaper when designed and constructed as piled foundations and such foundations are classified as deep foundations. (For piled foundation details see pages 272 to 292.)

Foundations are usually classified by their type such as strips, pads, rafts and piles. It is also possible to combine foundation types such as strip foundations connected by beams to and working in conjunction with pad foundations.

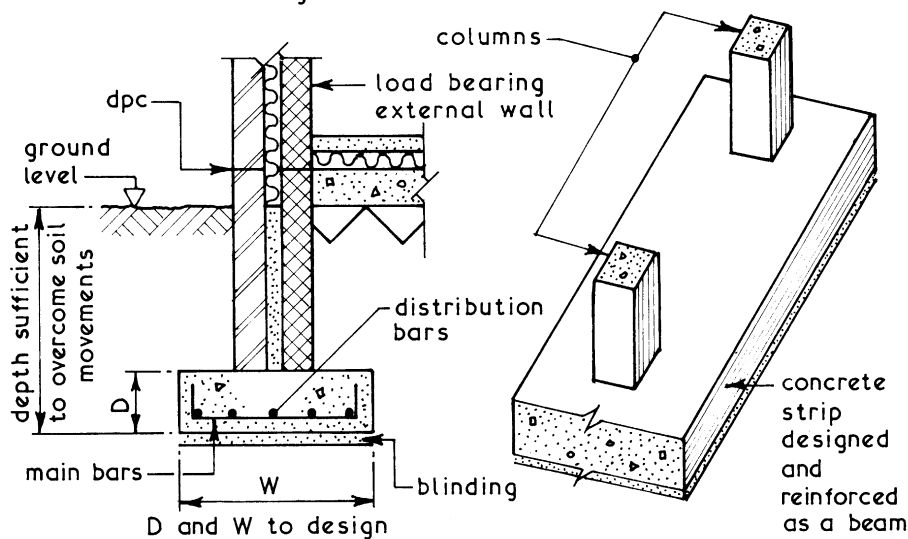
Strip Foundations ~ these are suitable for most subsoils and light structural loadings such as those encountered in low to medium rise domestic dwellings where mass concrete can be used. Reinforced concrete is usually required for all other situations.

Typical Strip Foundation Types ~



TRADITIONAL STRIP
low rise domestic dwellings
or similar buildings

DEEP STRIP or TRENCH FILL
alternative to traditional strip

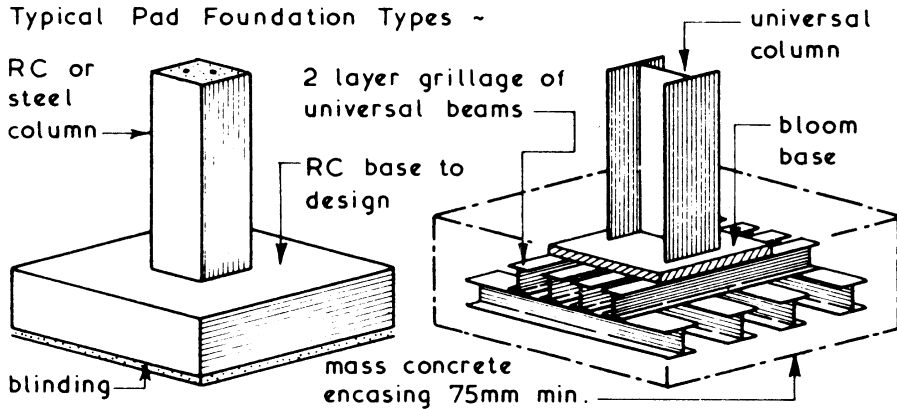


REINFORCED CONCRETE STRIP
used where induced tension exceeds
concrete's own tensile resistance

CONTINUOUS COLUMN
used for closely spaced or
close to boundary columns

Pad Foundations ~ suitable for most subsoils except loose sands, loose gravels and filled areas. Pad foundations are usually constructed of reinforced concrete and where possible are square in plan.

Typical Pad Foundation Types ~

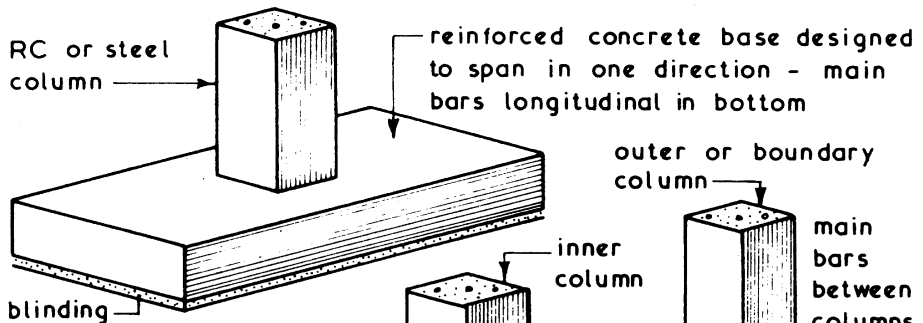


ISOLATED or PAD FOUNDATION

designed to span in two directions therefore main bars are placed in the bottom both ways

STEEL GRILLAGE

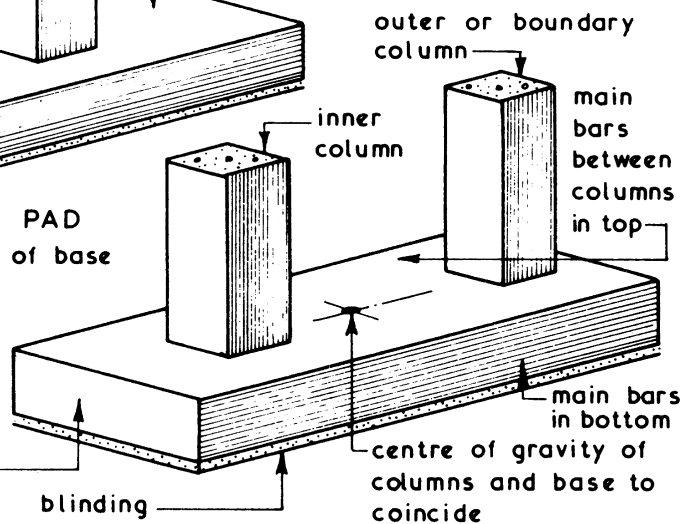
used for heavy concentrated loads - seldom employed today



RECTANGULAR PAD

used where width of base is restricted

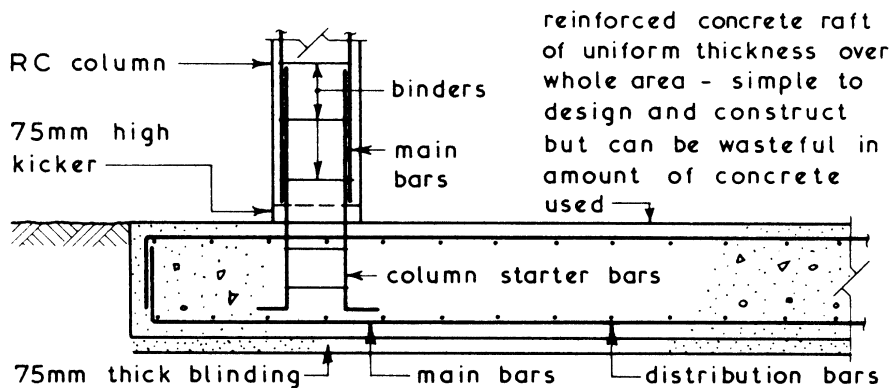
plan shape can be a rectangle or where length is restricted a trapezium



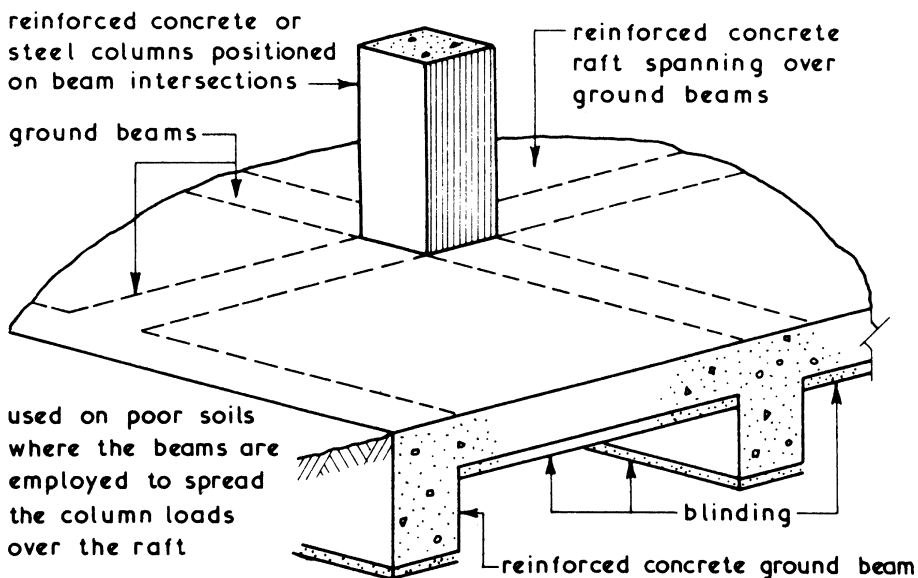
COMBINED COLUMN FOUNDATIONS - outer column close to boundary or existing wall

Raft Foundations ~ these are used to spread the load of the superstructure over a large base to reduce the load per unit area being imposed on the ground and this is particularly useful where low bearing capacity soils are encountered and where individual column loads are heavy.

Typical Raft Foundation Types ~



SOLID SLAB RAFT

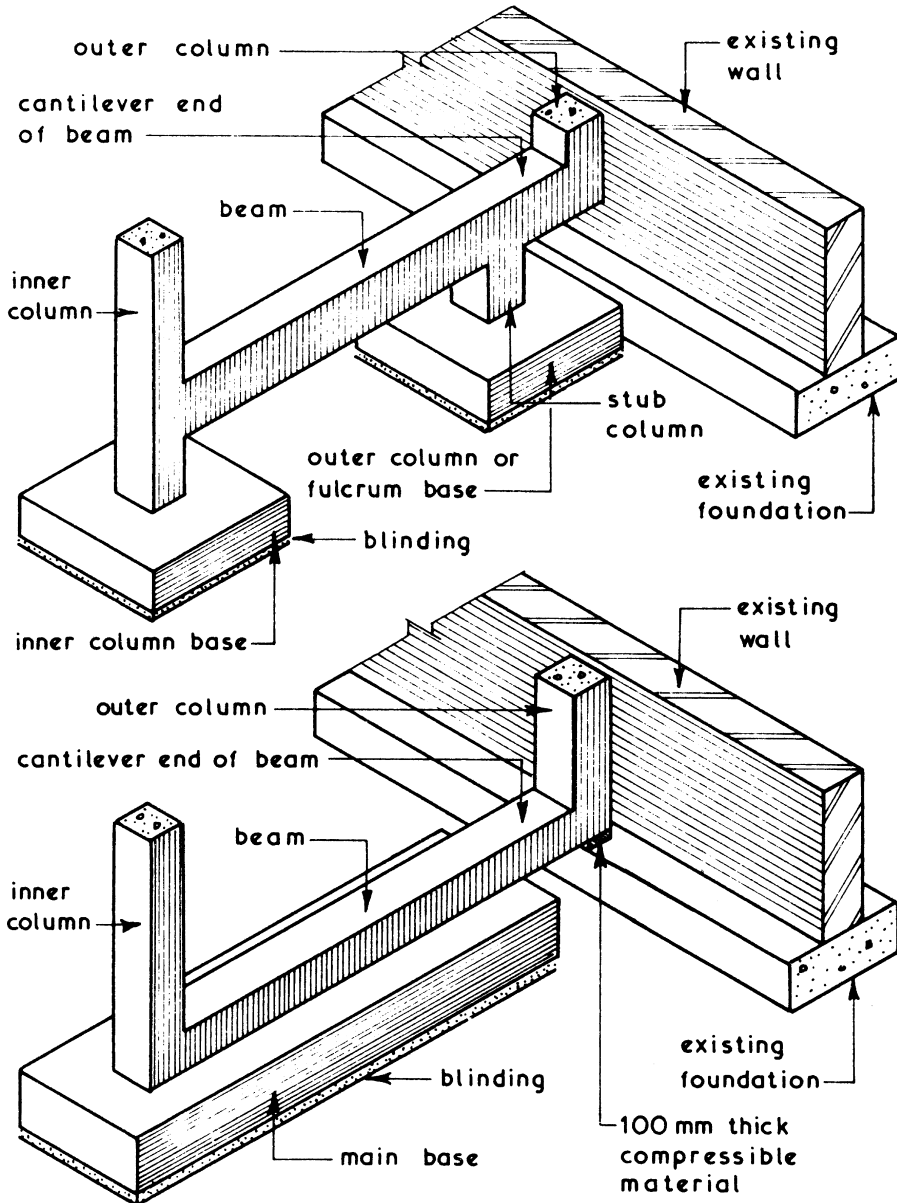


NB. Ground beams can be designed as upstand beams with a precast concrete suspended floor at ground level thus creating a void space between raft and ground floor.

BEAM AND SLAB RAFT

Cantilever Foundations ~ these can be used where it is necessary to avoid imposing any pressure on an adjacent foundation or underground service.

Typical Cantilever Foundation Types ~



Cantilever foundations designed and constructed in reinforced concrete